QC STORY PROJECT

Project Title:

"To eliminate the Defect of Steering column effort high for SC/DC Model ".

Mahindra and Mahindra Ltd,NASIK_ Plant, _SCO/XYLO TCF.

	Plan	Actual
Start Date	1.05.2019	31.5.2019
Completion Date	14.11.19	14.11.19

Current Status: **Completed**

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardize
- Step 9: Summaries & Plan future actions
- Guidelines to user

Team information

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	Jayesh Jirekar	Deputy Manager	Production	9860961900	Jirekar.jayesh@Mahindra.com
2	**Mentor**	Sandip chikhale	Sr Manager	Production	9922408138	Chikhale.sandip@mahindra.com
3	**Member 1**	Santosh Jadhav	Deputy Manager	MQA	9423303400	Jadhav.santosh@Mahindra.com
4	**Member 2**	Sanket Pimplapure	Deputy Manager	SQA	9503500998	Pimplapure.sanket@mahindra.com
5	**Member 3**	Prasad Deore	Senior Associate	Production	976405447	Deore.prasad@mahindra.com
6	**Member 4**	Pankaj patil	Deputy Manager	SQA	9011000369	Patil.pankaj@Mahindra.com

Step1. Select Subject

Brief Description :
To eliminate the Defect of Steering column effort high for SC/DC Model.

What : Steering column effort high.

Where : Vehicle rattle stage at CAI.

When : From May 19.

Who: By steering column up-down movement by Rattle stage associate..

Why: Chronic Repeat Concern, STR Loss, Quality loss, Cost loss, Deliver loss.

How: Avg. 25 cases / Month.

Steering column effort high

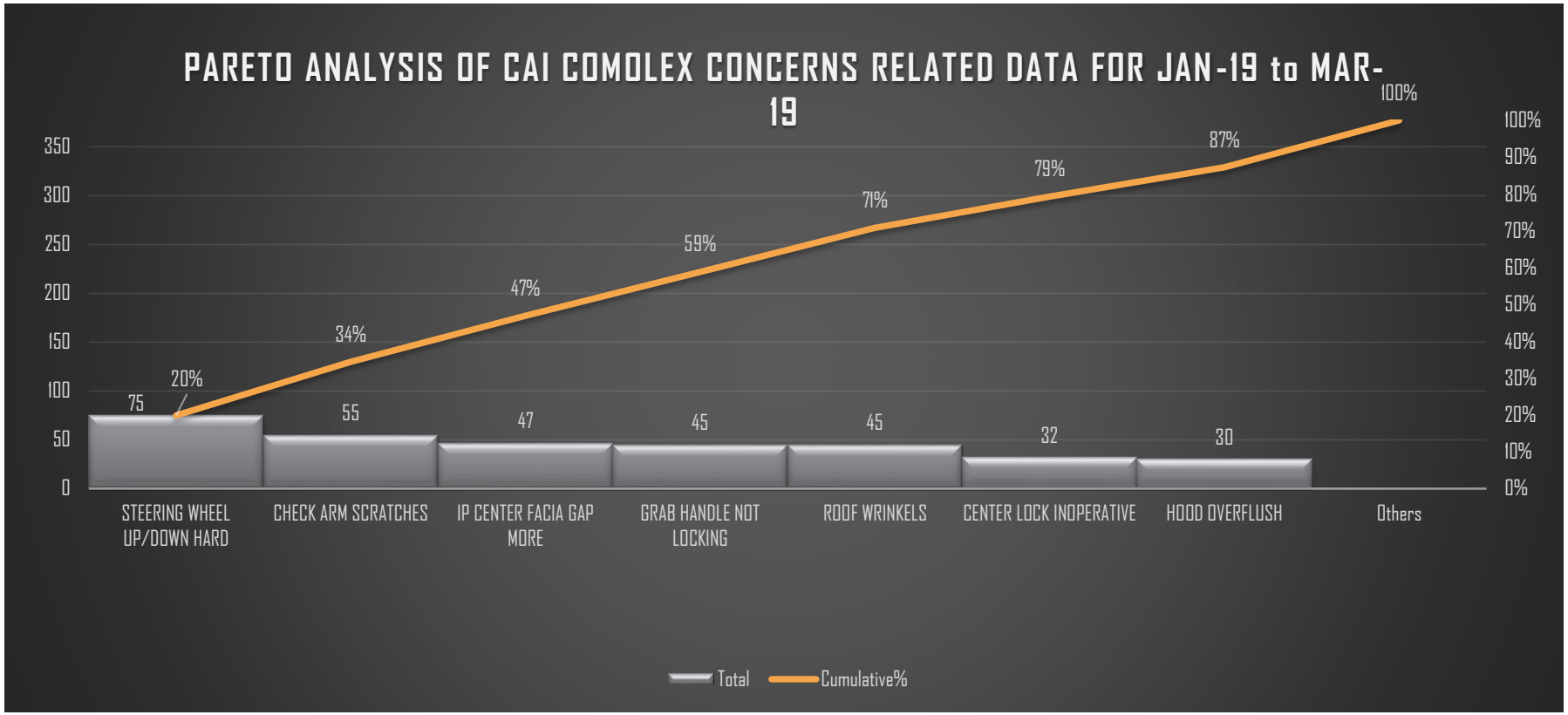
2. Justify the choice (reasons)

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- STR improvment

2. Justify the choice

- PARETO CHART ANALAYSIS FOR TRIM-1 DEFECT AT CAI



Inference :- Steering effort high is a chronic and time consuming defect at CAI .

2. Justify the choice (reasons)

Problem Criticality (Severity)

- **Importance & Severity of Concern :-** :

- **Steering column hard V1 concern.**
- **Major contributor in STR At CAI**
- **Avg. 25 cases reported in month**
- **Skill operator required for rework.**
- **Rework activity is critical in offline activity**
- **Repair cost is very higher RS <u>38206.5</u> per year .**

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	30
Avg Concerns/Day	Nos	1.0
Total Required Rework time per day	Min	30
Effective working time	Min	465
Operator required /day	nos	0.06
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	127.35
Avg no. of working days	days	25
Operator cost /month	Rs	3183.9
Operator Cost/year	**Rs**	**38206.5**

2. Justify the choice (reasons)

Problem Criticality (Severity)

- Steering column effort high, if this defect occurs it is resulted into highly customer dissatisfaction and it may cause sever accident during driving Severity of concern is 7 & 8

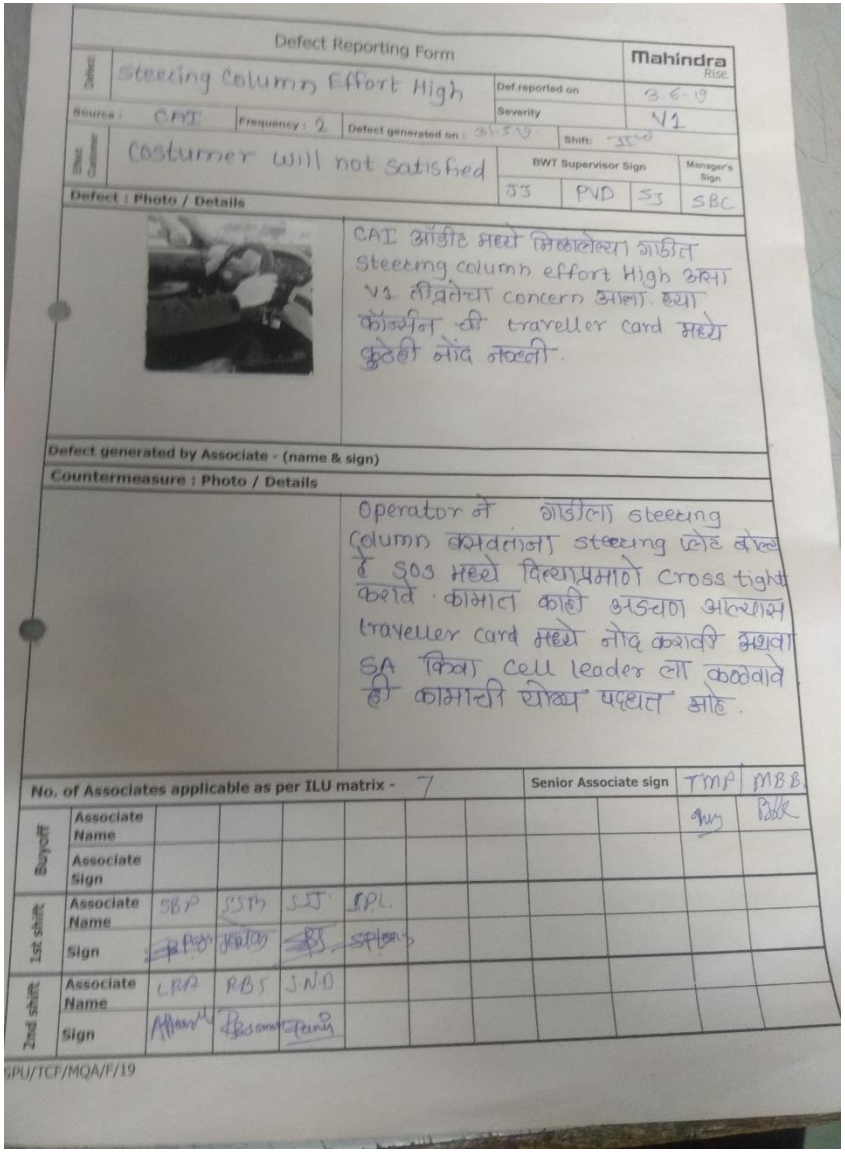
- Steering column effort high/ inoperative contributes leads to major rework.

- It creates Unsafe condition to next stage operation.

- Require 15 min to 20 min for rework the one vehicle.

2.b Immediate Containment Action

- Vehicle reworked at CAI rework area for the defect steering column up-down moment hard .
- Defect communicated to the stage operator through DRF.



Defect Reporting Form

Mahindra Rise

Defect: Steering Column Effort High

Def.reported on: 3.6.19

Source: CAI | Frequency: 2 | Defect generated on: 31.5.19 | Severity: V1

Shift: 1st

Effect Customer: Costumer will not satisfied

BWT Supervisor Sign			Manager's Sign
JJ	PVD	SJ	SBC

Defect : Photo / Details

CAI ऑडीट मध्ये निघालेल्या गाडीत Steering column effort High असा V1 तीव्रतेचा concern आला. ह्या कॉलम्स ची traveller card मध्ये कुठेही नोंद नव्हती.

Defect generated by Associate - (name & sign)

Countermeasure : Photo / Details

Operator ने गाडीला steering column कसतांना steering व्हील बोल्ट ने SOS मध्ये दिल्याप्रमाणे cross tight करावे. कामात काही अडचण असल्यास traveller card मध्ये नोंद करावी अथवा SA किंवा cell leader ला कळवावे ही कामाची योग्य पद्धत आहे.

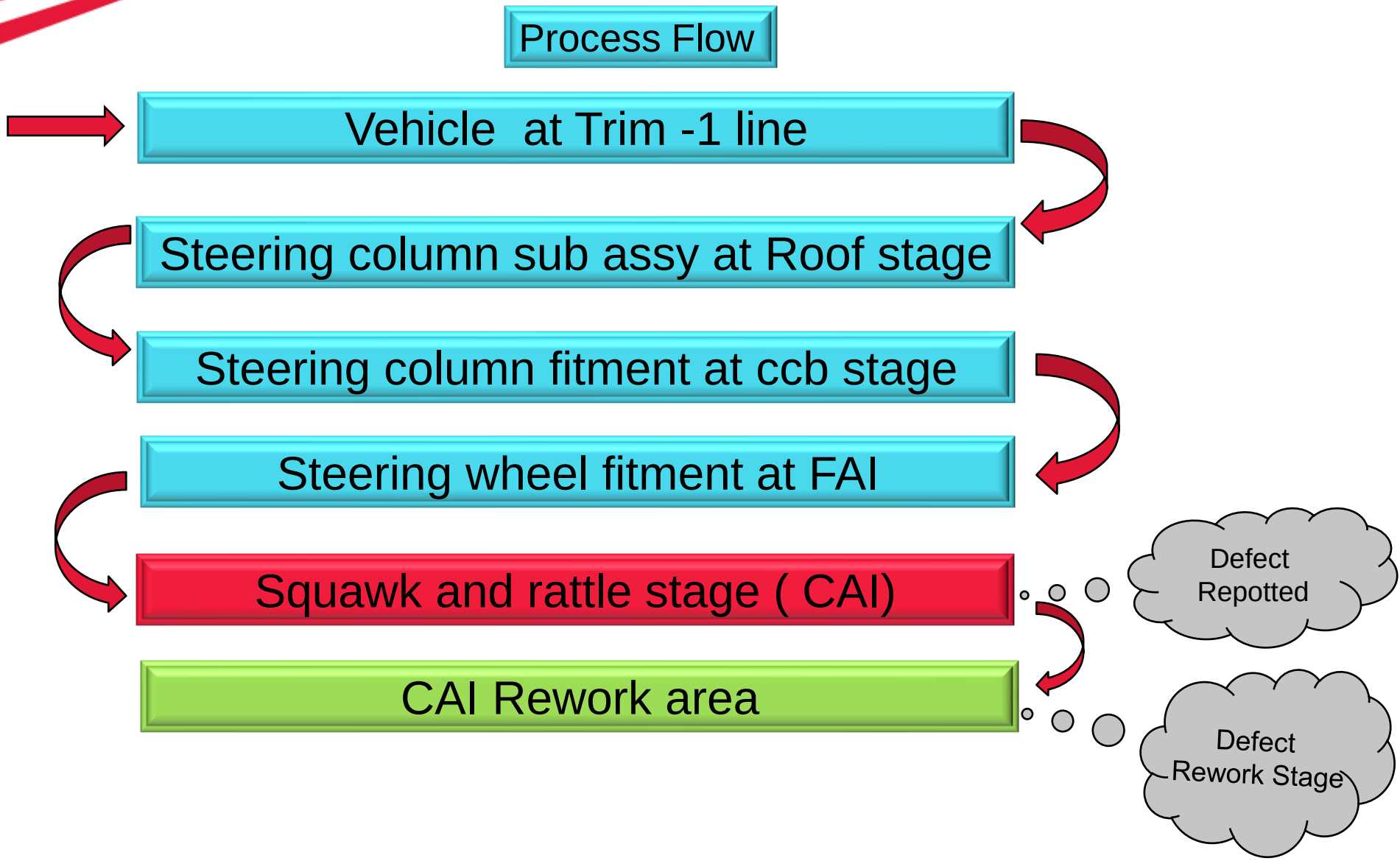
No. of Associates applicable as per ILU matrix - 7

Senior Associate sign: TMP, MBB

Buyoff	Associate Name										
	Associate Sign										
1st shift	Associate Name	SBP	SSM	SJ	SPL						
	Sign										
2nd shift	Associate Name	LRA	RBJ	JND							
	Sign										

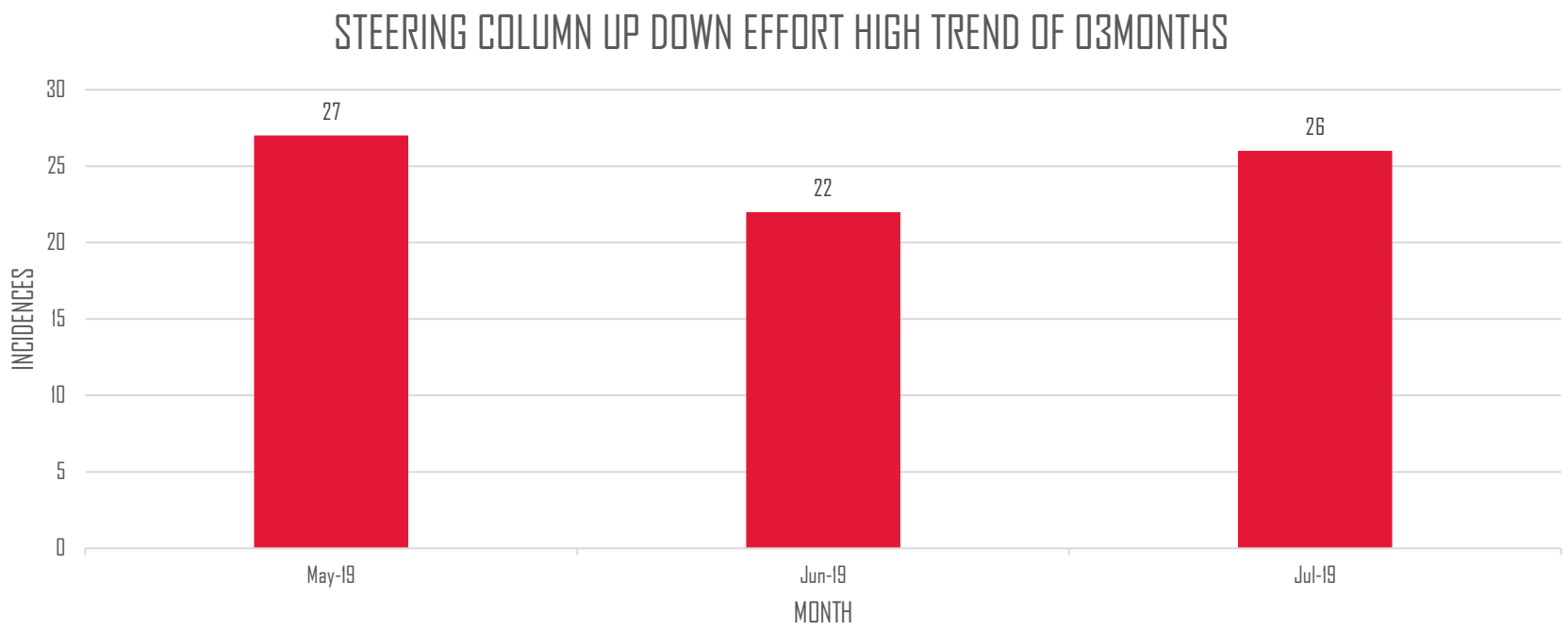
GPU/TCF/MQA/F/19

3. Understand the current situation



3. Understand the current situation

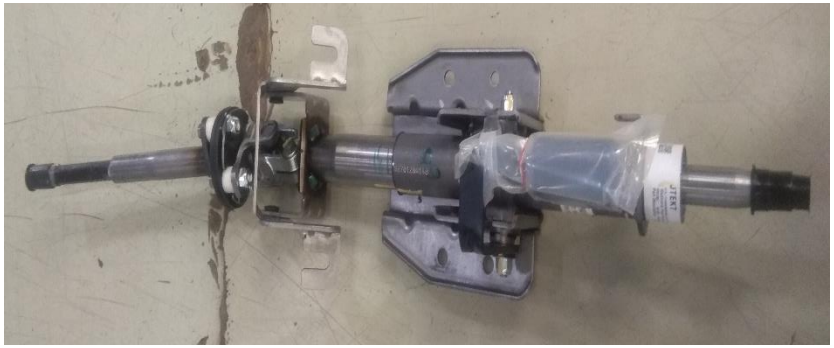
Month wise Steering column effort high concern Trend at CAI for May 19- to Jul-19



Inference – Month on month defect trend is increasing

3. Understand the current situation

Parts used For Steering column fitment.
(2 Types parts and 2 Types Hardware)



Steering column.

CCB.

Bolts .

Nuts

3. Understand the current situation

Operation.

Steering column UJ insert in grommet

Pre-fitment of bolt and nuts on Steering column.

Tightning of bolt & Nuts on CCB.

3. Understand the current situation

4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area : Scorpio/Xylo TCF
Cell / BWT / Line: Trim2/Bwt 2

Requested By- MQA | Date -03- 06-19 | Source -CAI | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement : Steering column effort high

FMEA Severity /Gravity : ☐ V1+ ☐ V1 ☑ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 1 | VIN / Aggre. No.- | Model : Scorpio

Date of Incidence/ Assy :31.5.19 | Shift / Time : 2nd | Whether SC/CC/Z parameter affected ? ☑ YES ☐ NO ☐ Z ☑ CC ☐ SC

Immediate containment action for Customer Protection :
Yard check done

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. / QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned to - | BWT . :

Does a control already exist ? ☐ YES ☐ NO

Stage / Post Analysis -

Man(Human Action)

Operator Name : L R Attarde
Is operator trained ? ☑ YES ☐ NO
Operator level (LL.U) : L
OR Dexterity level(1…4) : 3
Is operator under training ? ☐ YES ☑ NO
Is operator temporary for Stage ? ☐ YES ☑ NO
Bad rework ? ☐ YES ☑ NO

Method(process)

Main steps are respected ? ☑ YES ☑ NO
Is there a related key point in SOP / SOS? ☑ YES ☐ NO
Are the reasons for key point known ? ☑ YES ☐ NO
Are operating reqmnts respected ? ☑ YES ☐ NO
Is the defect part of the buy off / Checkman list ? ☐ YES ☑ NO
Was it detected by Inspector/ Buoff Opr ? S B Sawant ☐ YES ☑ NO

Other

Is there a Control Plan ? (Record in Analysis table) ☐ YES ☐ NO
Is the packaging standard defined for the part ? ☐ YES ☐ NO
Is the packaging standard respected ? ☐ YES ☐ NO
Is there any error mechanism for failure ? ☐ YES ☐ NO

Machine

Is the specified /Tool / Fixture facility used ? ☐ YES ☐ NO
Last calibration dates
Facility /Tool / Fixture in good condition ? ☐ YES ☐ NO
Is it a substitute facility ? ☐ YES ☐ NO
Are machine parameters known ? (Record in Analysis table) ☐ YES ☐ NO
Parameters are respected ? ☐ YES ☐ NO

Product Failure

In only one shift ? ☐ YES ☑ NO
In only one line ? ☐ YES ☑ NO
The problem follows the vehicle ? ☑ YES ☐ NO
The problem follows the part ? ☑ YES ☐ NO
Is it a repeat problem ? ☑ YES ☐ NO
Is the suspected parameter measurable ? (Record in Analysis table) ☐ YES ☐ NO

Material/ Consumable

Is the specified product used ? ☐ YES ☐ NO
Did it evolve / Change ? ☐ YES ☐ NO
Is the storage SOP respected ? ☐ YES ☐ NO
Is the shelf life / expiry date respected ? ☐ YES ☐ NO
Is the part as per FIFO? ☐ YES ☐ NO

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusionon Gemba (Attach Sketch/ Photograph if necessary) -
During analysis we observed that, steering column effort high vehicles reported at rattle stage in cai, in that case we have loose and thighten the nut and bolts of steering column as per sos still concern remain same. in further analysis we observed that, We have deployed Mr. L R attarde on steering column fitment stage, he has fitted stering column as per sos also he followes the tightning sequence.

Sno	Product / Process / Machine / Parameter	Spec.	Actual	Remark
1				

Is 4M Analysis Part 2 required ? ☐ YES ☑ NO

Is the substitute transport mode used ? ☐ YES ☑ NO **Is the packaging defined ?** ☐ YES ☐ NO

Actions -

1.Containment Action	Resp	Date	Status
1.DRF given to all stage associate and buyoff checkman	JJ/PVD	3.6.19	closed

2. Permanent Action -	Resp	Date	Status

Analysed by Name/s - Jayesh jirekar
Date -3.6.19

Choice : ☐ WSS ☐ QC Story

4M analysis Good and Not so good observations

MAN	Main step are respected by associate.
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	Problem follows part
Material	Handling and shelf life of part adhered

3. Understand the current situation

4M Analysis Observation.

- Concern Reported at CAI squeak and rattle stage.
- **All L3 Associates Working** on Stage.
- Concern Reported in Scorpio vehicle **all models .**
- Concern reported as Steering wheel up down effort high.
- **Rework** is done at the rework area at CAI.
- **Rework** Takes more than 1/2 **Hrs** time if bolt/nut get sli[p.

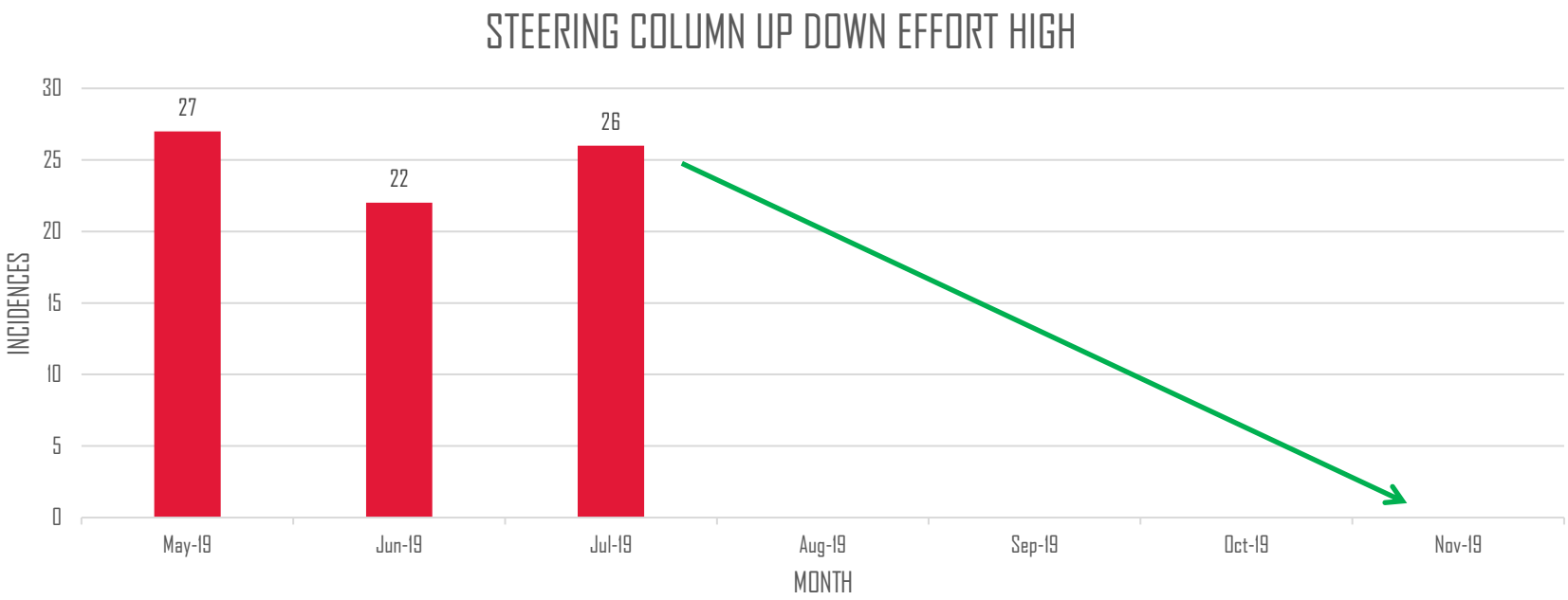
4. Select targets

Present Level:

Average 25 Defects/Month

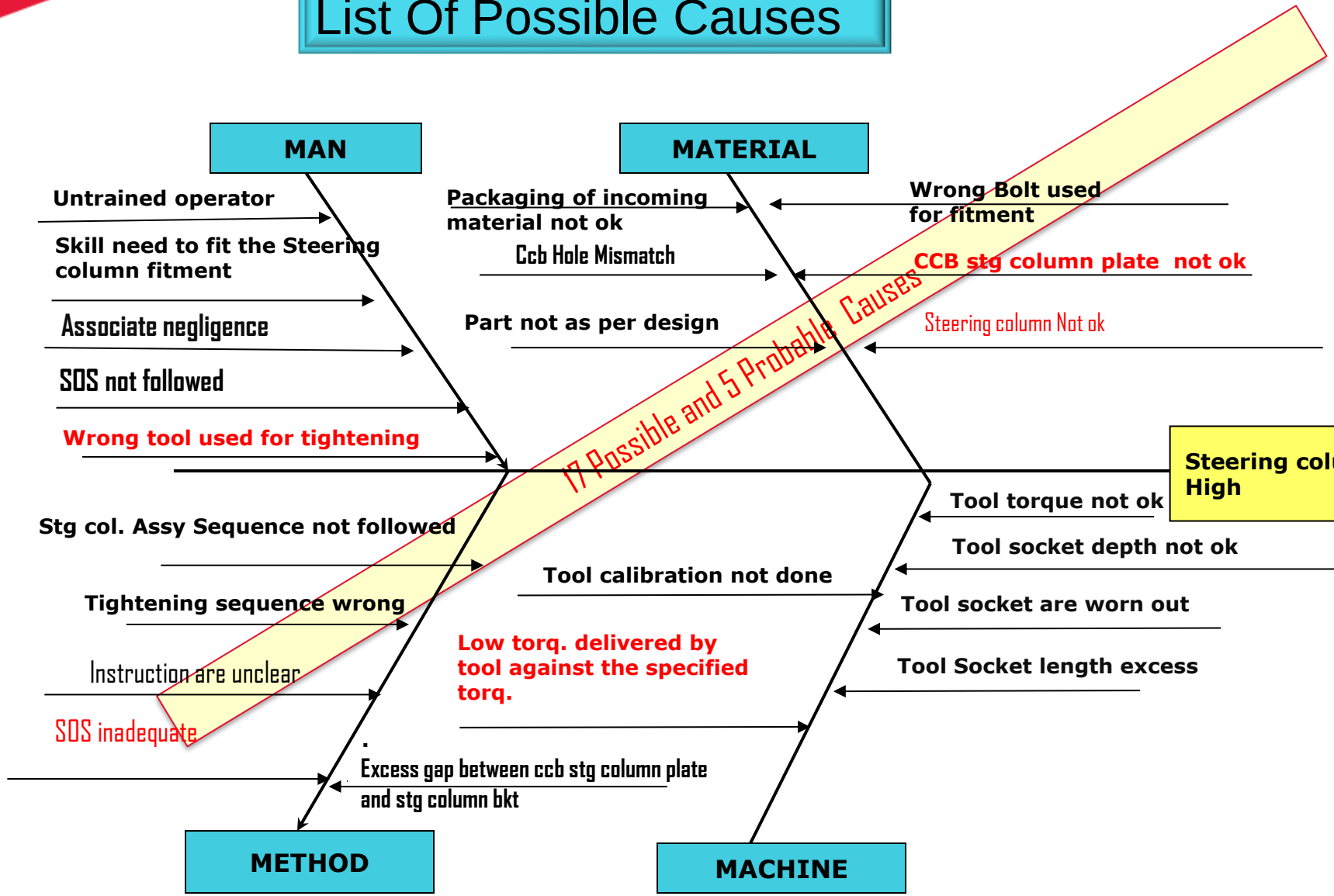
Target:

To Eliminate defect of steering column effort high by Nov 2019



Step 5. Analysis

List Of Possible Causes



5. Analysis

5.1 Identify possible causes: Brainstorming

SR NO	POSSIBLE CAUSES	Jayesh	Amar	Santosh	Pankaj	Sanket	TOTAL
1	Untrained operator	1	1	1	1	1	5
2	Skill need to fit the Steering column fitment	3	3	3	3	1	13
3	Associate negligence	3	3	3	3	1	13
4	SOS not followed	1	1	1	9	1	13
5	Wrong tool used for tightening	9	3	9	1	9	31
6	Stg col. Assy Sequence not followed	3	3	3	3	1	13
7	Tightening sequence wrong	3	1	1	1	3	9
8	Instruction are unclear	3	1	1	1	3	9
9	Ccb Hole Mismatch	3	1	1	1	3	9
10	Part not as per design	3	3	3	3	3	15
11	Tool calibration not done	3	3	3	3	3	15
12	Low torq. delivered by tool against the specified torq.	3	9	9	9	3	33
13	Excess gap between ccb stg column plate and stg	3	3	3	3	3	15
14	Wrong Bolt used for fitment	3	3	3	3	1	13
15	Steering column Not ok	9	3	9	1	9	31
16	SOS inadequate	9	3	3	9	9	33
17	Tool torque not ok	3	3	3	3	3	15
18	Tool socket depth not ok	3	3	3	3	3	15
19	Tool socket are worn out	3	3	3	3	3	15
20	Tool Socket length excess	3	3	3	3	3	15
21	CCB stg column plate not ok	9	3	9	1	9	31

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting cause for hypothesis is 31.

Step 5. Analysis

5.1 List of Possible causes: Brainstorming

List of causes not considered as probable causes and reasons for not considering

Sr.N	Possible causes	Reasons
1	Untrained operator	Trained associate deployed at stage
2	Skill need to fit the Steering column fitment	L3 level dexterity associate deployed at stage
3	Associate negligence	No associate neglygency observed.
4	SOS not followed	SOS followed by stage associate.
5	Tightening sequence wrong	check found ok
6	Stg col. Assy Sequence not followed	Associate followed sub assy sequence
7	Instruction are unclear	Instruction clear mention in SOS
8	Part not as per design	Part as per drawing
9	Tool calibration not done	Tool calibration done as per shedule
10	Wrong Bolt used for fitment	bolt found as per specification and drawing
11	Ccb Hole Mismatch	Part checked found ok
12	Excess gap between ccb stg column plate and stg	Part checked found ok
13	Tool torque not ok	Tool torque found ok
14	Tool socket depth not ok	Tool socket depth found ok
15	Tool socket are worn out	No socket wear out
16	Tool Socket length excess	Tool socket length found ok
17	Packaging of incoming material not ok	Packaging as per packaging standard.

5. Analysis

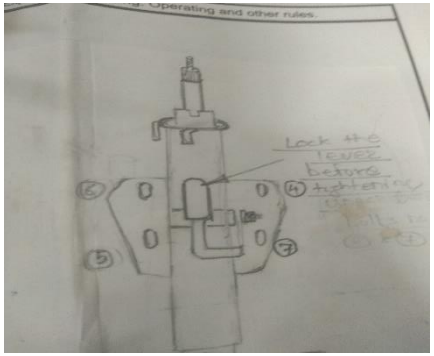
5.1 List of Probable causes: Brainstorming

Sr No	Probable cause
1	SOS inadequate
2	Steering column Not ok
3	Low torq. delivered by tool against the specified torq.
4	CCB stg column plate not ok
5	Wrong tool used for tightening

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	SOS inadequate	1) Visual	CFT member observe the stage for 2 hrs. Observations are as below… 1. SOS followed by stage associate still defect occur in some vehicle .	**Hypothesis valid**



Existing tightening sequence

10 Vehicles checked for steering column effort high out of which 3found not ok

Sr.No	Vin No	Remark
1	49022	ok
2	49028	ok
3	49112	not ok
4	49120	ok
5	49125	ok
6	49128	ok
7	49130	not ok
8	49134	ok
9	49137	not ok
10	49138	ok

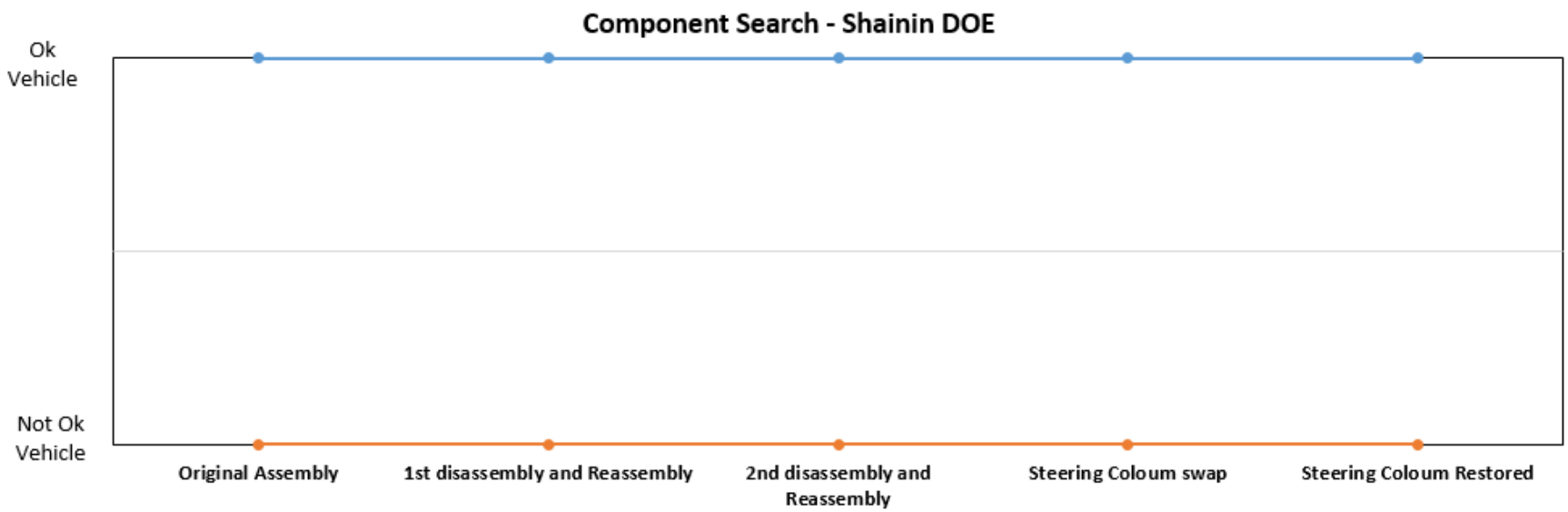
Concern	Stg col effort ok	Stg col effort high
Vehicles checked	7	3

Trim-1 line it is observe that After following srg column tightening sequence some vehicles get not ok

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	Steering column not ok	DOE	We have changed steering column for defect of effort high but after changing steering column defect remain same	Hypothesis invalid



Shainin DOE confirms – Defect does not followed to Steering Colum hence it is not a cause

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

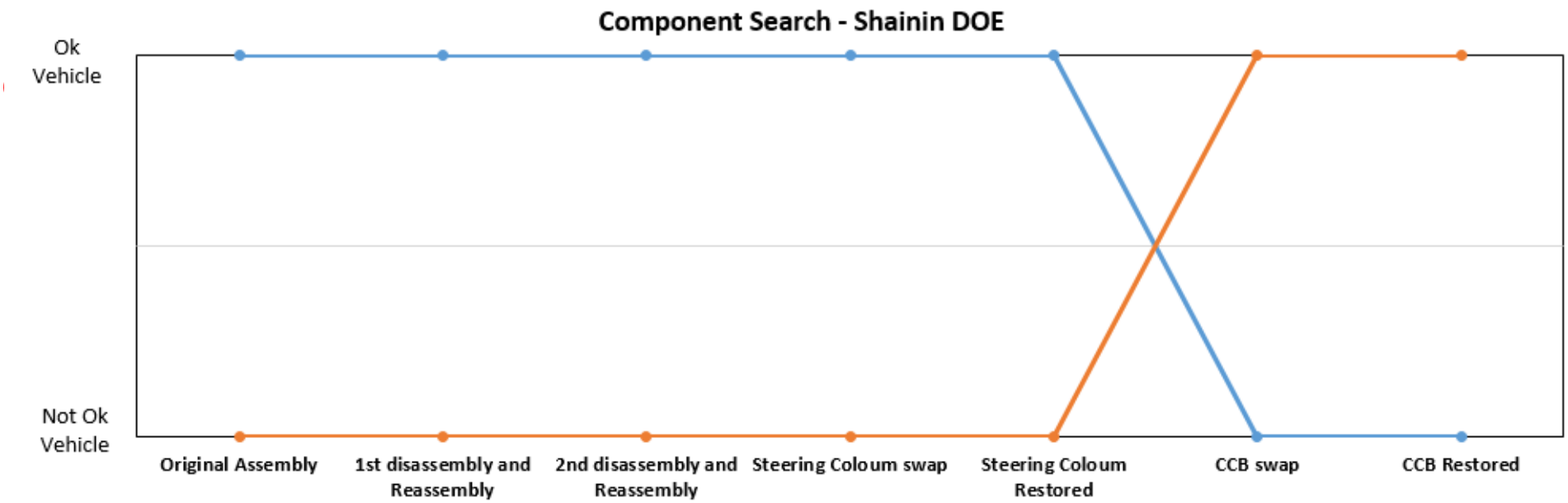
No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Low torq. delivered by tool against the specified torq.	Acta m/c	15 vehicles checked for torque , we observed that torque within limits.	**Hypothesis Invalid**

Sr No	Torque Value	Remark
1	22	ok
2	21.9	ok
3	21	ok
4	98	ok
5	21.98	ok
6	21.85	ok
7	21.88	ok
8	22.1	ok
9	22	ok
10	22.2	ok
11	22	ok
12	21.99	ok
13	21.99	ok
14	21.95	ok
15	21.94	ok

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	CCB stg column plate not ok	DOE	During observation it is observe that ccb of stg col. Effort high vehicle ccb changed and defect was not observed.	**Hypothesis valid**



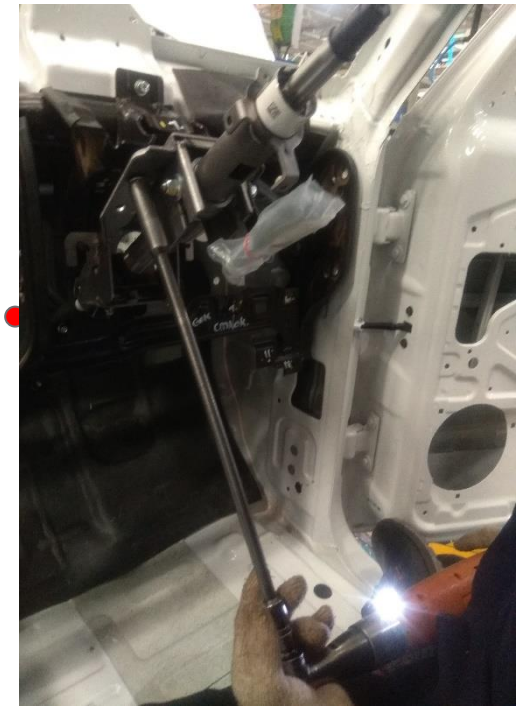
Shainin DOE confirms – Defect followed to CCB hence it is a cause

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	Wrong tool used for tightening	1) Visual	We have checked 15 vehicles for correct tool used by stage associate , we found that all vehicles tighten by correct tool	**Hypothesis invalid**

We have checked 15 vehicles for correct tool used by stage associate , we found that all vehicles tighten by correct tool



Sr. No.	Veh Vin No	Remark
1	49100	ok
2	49103	ok
3	49104	ok
4	49105	ok
5	49106	ok
6	49107	ok
7	49116	ok
8	49112	ok
9	49134	ok
10	49138	ok
11	49125	ok
12	49131	ok
13	49137	ok
14	49099	ok
15	49108	ok

Observation :- we have checked 15 vehicle for correct tool used foud ok.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	SOS inadequate	Visual	Observations are as below… 1. SOS followed by stage associate still defect occur in some vehicle	Hypothesis valid
2	CCB stg column plate not c	DOE	During observation it is observe that ccb of stg col. Effort high vehicle ccb changed and defect was not observed.	Hypothesis valid

2 Valid Causes

5.3 Identify the root causes:

WHY - WHY Analysis

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTERMEASURE
1	SOS inadequate	Tightning sequence of steering col fitment not effective	Vehicle not ok after tightening sequence followed	Tightening sequence incorrect	Change the Tightening sequence
2	CCB stg column plate not ok	Lower bkt forming bottoming was observed in variation	Variability at supplier end not detected at supplier	No checking provision	Implement the checking gauge

6. Implement the corrective measures

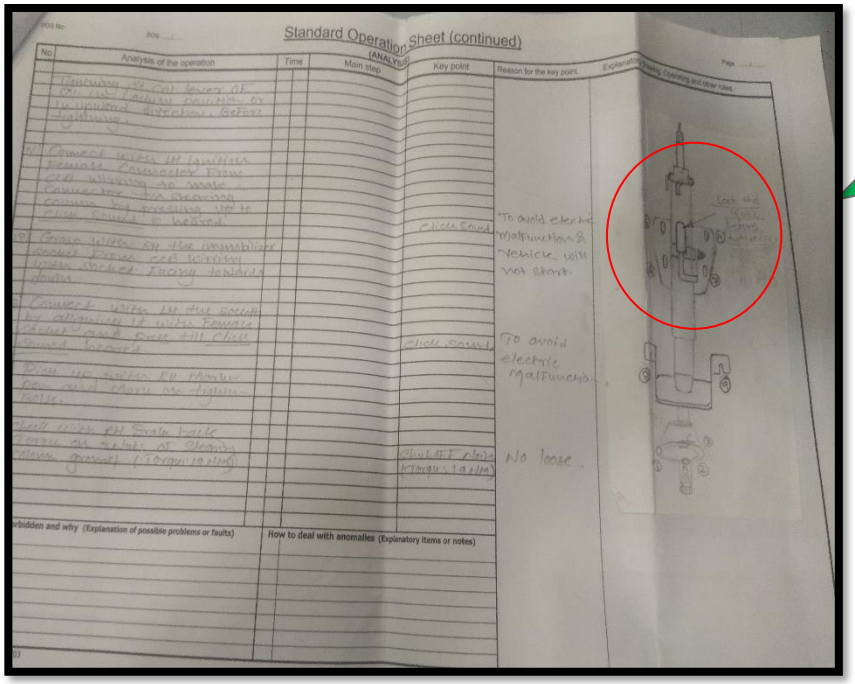
6.1 Actions Decided

No	Root Cause	Action plan	Target date	Responsibility	Status
1	Tightening sequence incorrect	Change the Tightening sequence	12.8.19	Jayesh Jirekar	Closed
2	No checking provision	Implement the checking gauge	19.9.19	Pankaj Patil	Closed

6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
1	Tightening sequence incorrect	Change the Tightening sequence of stg column	12.8.19	Jayesh Jirekar	Closed

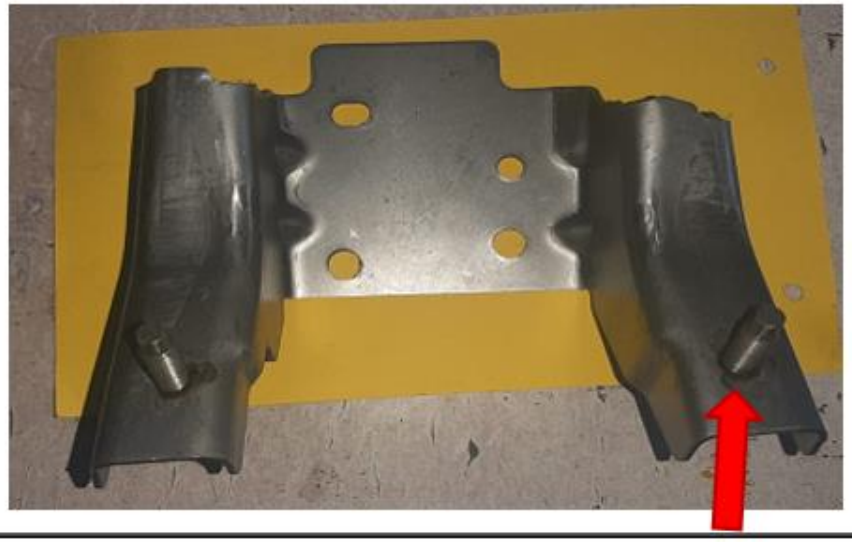
Sr no	Veh Vin No	Trial taken with new tightning sequence
1	25443	Found ok
2	25446	Found ok
3	25449	Found ok
4	25450	Found ok
5	25459	Found ok
6	25475	Found ok
7	25486	Found ok
8	25489	Found ok
9	25482	Found ok
10	25502	Found ok



Stg column tightening sequence changed.

6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
2	No checking provision	Checking gauge implemented to check the flatness checking	19.9.19	Pankaj Patil	Closed

BEFORE	AFTER
	

Problem: Lower brkt projection inprocess stage Visual inspection

Improvement: Handy gauge using started at Projection welding inprocess stage for gauging & flatness checking.

6. Implement the corrective measures

CCB Supplier SCAP Report

Microsoft Excel 97-2003 Worksheet

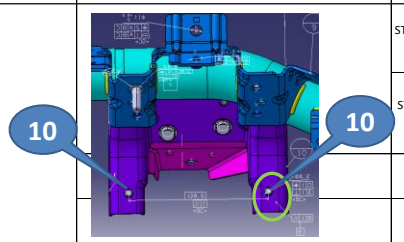
Supplier SCAP Report

TRACK COMPONENTS LTD

FINAL INSPECTION QUALITY PLAN [F I Q P]

19.09.19

Part Name : ASSY CROSS CAR BEAM LHD | **Part No. 0112AAL00300N**

S.NO.	CATEGORY		Parameter			Specification	Insp. Method	Freq.
1	FINISH		No rusty of part , No rusty of nut & Stud , No sharp edges , No Bend			Ok / Not Ok	Visual	100 % / Lot with dot
2	FITMENT	No. of Nut & Bolt	No of M6 Nut			13 Nos	Visual	100 % / Lot with sr. no.
3			No of M8 Nut			8 Nos	Visual	100 % / Lot with sr no.
4			No of M8 Stud			3 Nos	Visual	100 % / Lot with dot
5			No fo M5 Stud			2 Nos	Visual	100 % / Lot with dot
6			Stering column movement jam				Handy Gauge	100 % / Lot with dot
7			M8 Nut Thread Size & no mismatch & no spatter in nut			Ok / Not Ok	M8 x 1.25 Thread plug gauge	100 % / Lot with dot
8			STEERING COLUMN MOUNTING STUD POSITION [W.R.T. SURFACE ABC]	SC		Ok / Not Ok	HANDY RELATION CHECKING GAUGE	100 % / Lot with dot
9			STEERING COLUMN MOUNTING INDIVIDUAL STUD PERPENDICULARITY [W.R.T. SURFACE E]	SC		Ok / Not Ok	HANDY RELATION CHECKING GAUGE	100 % / Lot with dot
10			Steering Column mtg.lower Brkt Surface //SM SURFACE E w.r.t. D			1 .0 MaX	R Checking gauge & Filler Gauge (Go-0.5 & No-go-1.0mm)	100 % / Lot with dot
11			Manual Assy. Of Plenum lower Mtg. brkt DS			Pre-tightening of M8 bolt - 2nos	Manual & Visual	100%

PREPARED BY
Pankaj Patil

APPROVED BY
RAVINDRA PUNNASE

Format No:QF/QA/42 Rev No:02 Rev Date:01.04.2018

7. Check & confirm the results & effects

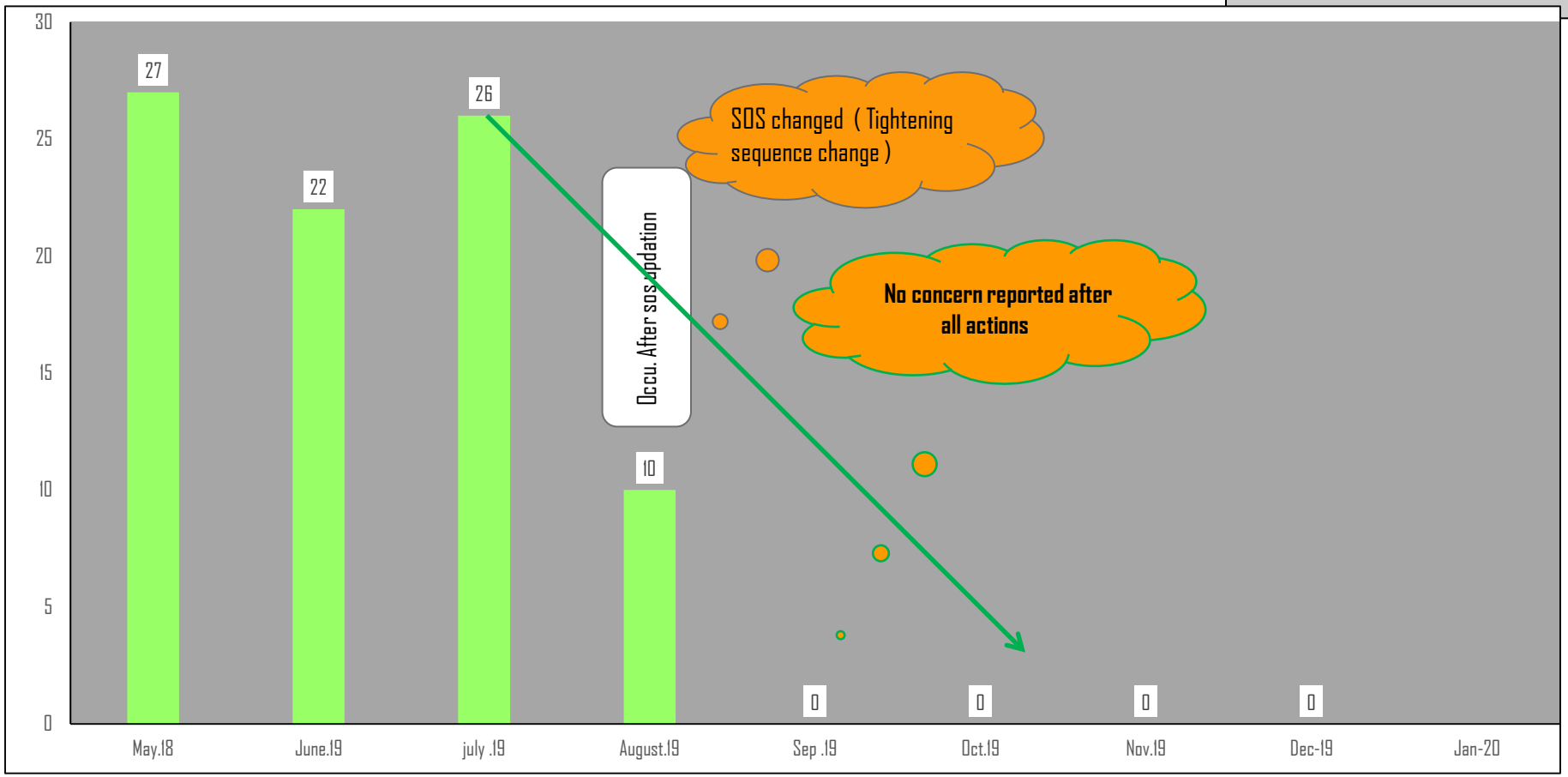
7.1 Verify Results:

S N	Action Taken	Testing Method	Results Before Action	Results After Action
1	Change the Tightening sequence of stg column	Process audit sheet	Stg column effort high due to SOS adequate	Stg column effort high defect reduced due to correction in soe (Tightening sequence)
2	LHD lower brkt Forming tool correction & blue matching to be done.	Verifying of part with drawing	Stg column effort high due to ccb part quality issue	No issue observe after correction in ccb

7. Check & confirm the results & effects

7.1 Verify Results: Defect Trend

X Axis- Month
Y Axis- cargo bolt not fitted .



Inference - Concern is eliminated after all 3 actions implemented.

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Target date	Responsibility	Status
1	Change the Tightening sequence of stg column	12.8.19	Jayesh jirekar	Closed
2	LHD lower brkt Forming tool correction & blue matching to be done.	19.9.19	Pankaj Patil	Closed

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Target date	Responsibility	Status
1	SOS Updated(tightening sequence changed.	12.8.19	Jayesh Jirekar	Closed
2	OPL	12.8.19	Jayesh Jirekar	Closed
3	Scap report.	19.9.19	Pankaj Patil	Closed

Check & confirm the results & effects (Cont..)

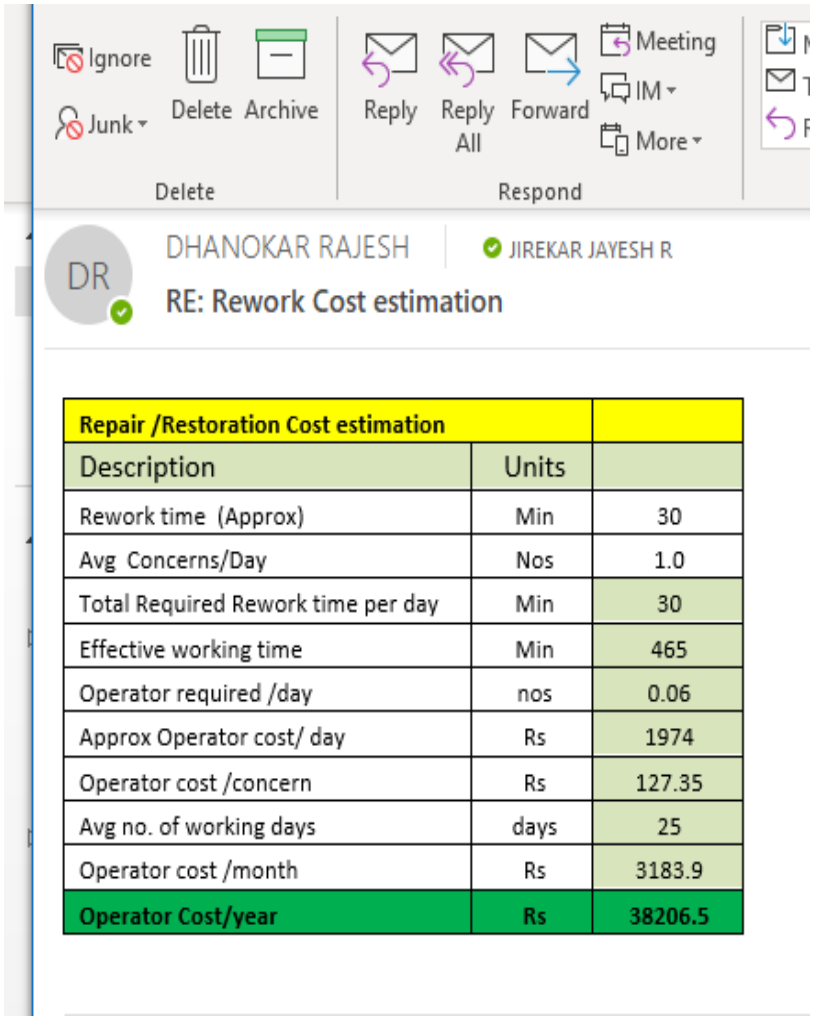
7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) Chronic concern for STR loss is eliminated.
2) COPQ (Material cost+ Labour cost+ Rework cost) =(Apprx. Rs.38206.5 Per annum)is Saved

In tangible Benefits

1) Associate fatigue eliminated.
2) Improved Customer Satisfaction.



Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	30
Avg Concerns/Day	Nos	1.0
Total Required Rework time per day	Min	30
Effective working time	Min	465
Operator required /day	nos	0.06
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	127.35
Avg no. of working days	days	25
Operator cost /month	Rs	3183.9
Operator Cost/year	**Rs**	**38206.5**

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Target date	Responsibility	Status
1	SOS Modification	19.9.19	Santosh Jadhav	Closed
2	OPL	12.8.19	Jayesh Jirekar	Closed

8.2 Training:

SN	Training Identified	Target date	Responsibility	Status
1.	OPL given to stage associates	25.8.19	Jayesh jirekar	Closed

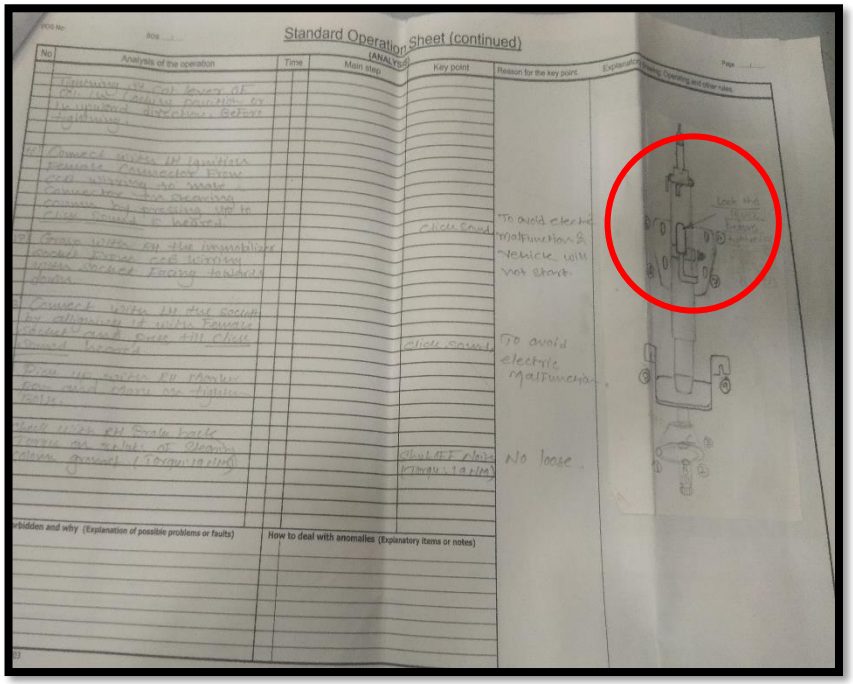
8.3 Additional Audit / Check

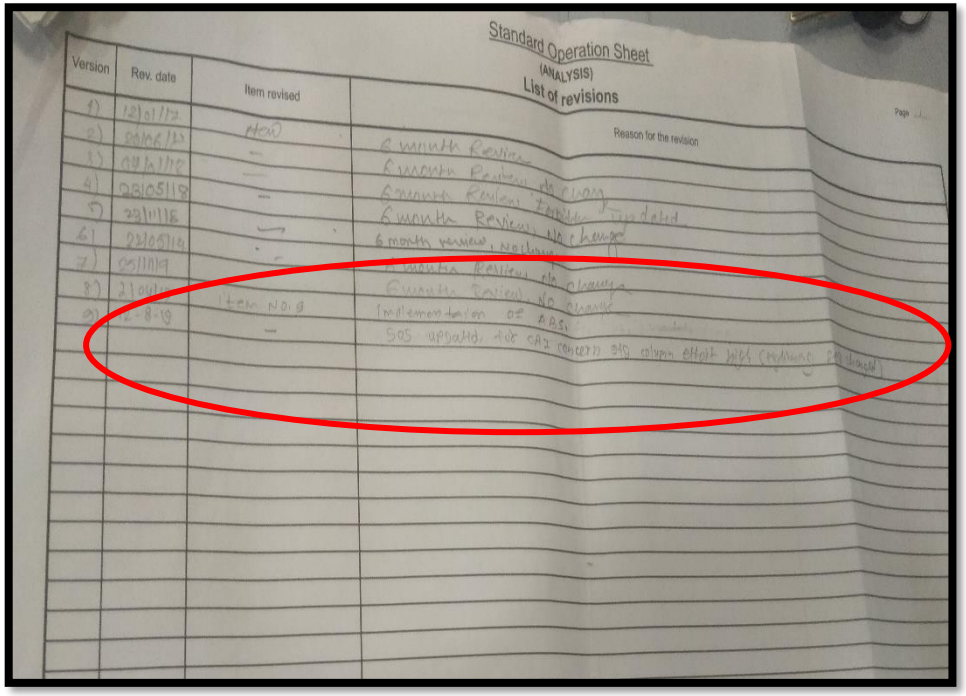
1	Check point at supplier end	19.9.19	Pankaj Patil	Closed

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SOS





8. Standardization

8.2 Training

2	OPL given to stage associates	12.08.19	Jayesh	Closed

8. Standardization

8.3 Additional Audit / Check

2	Check point at supplier end	19.9.19	Pankaj Patil	Closed

QUALITY ALERT _ FOR INSPECTION

COMPLAINT DATE: 19.09.2019

QTY : 1 NO.

CUSTOMER :- Mahindra and Mahindra. LTD.

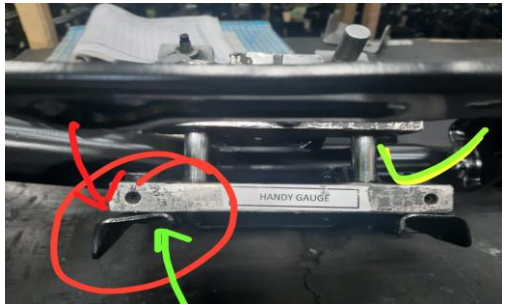
PART NO. : 0112AAL00300N

PART NAME :-:ASSY CROSS CAR BEAM -S201

REASON FOR QUALITY ALERT.

Stering column movement jam observed at Scorpio TCF.(Excess gap observed at lower brkt)

DON'T [NOT OK PART PHOTO]



Checking Control Method :

No Excess gap allowed at lower brkt check with filler gauge.

DO'S [OK PART PHOTO]

100‰ cale k r Naova paT - var iT k nak - k r Nao

9. Summaries & Plan future actions

9.1 Conclusion:

Steering column effort high defect is eliminated after

- Correction done in CCB .
- Tightening sequence change .

9.2 Horizontal Deployment:

NA

9.3 Lessons Learnt:

Things Gone Wrong :

Such type of defect not consider at starting phase

Things Gone Right : .

Small correction in tightening sequence has given the good improvement.

Thank You.....